

PROJECT DESIGN PHASE

Solution Architecture

Date	01 October 2026
Team ID	SWTID-2026-9397
Project Name	Auto-Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Solution Architecture

The solution architecture describes how the business requirement for simpler school IT ticket submission is connected to the ServiceNow technology components. The design separates the user interface, ticket data, automation logic, notification service, and controlled configuration.

Multi-Tier Solution Architecture Flowchart

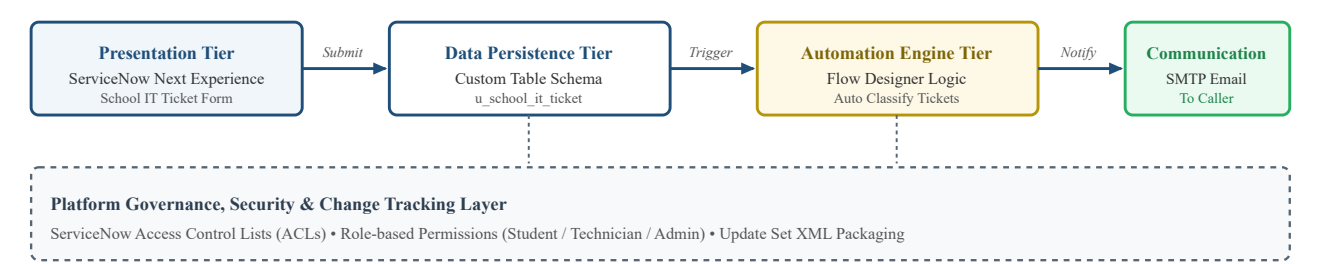


Figure 1: Multi-tier solution architecture connecting UI, schema, Flow Designer, and platform services.

Architecture Flowchart & Diagram Explanation

System Flow & Layer Interaction:

- **Presentation Layer:** Students interact with the ServiceNow Next Experience School IT Ticket form. They input only their identity (Caller) and issue summary (Short Description).
- **Data Persistence Layer:** The custom table u_school_it_ticket captures the record and triggers database events.
- **Automation Layer:** Flow Designer listens asynchronously for record creation, inspects text fields, evaluates conditional keyword rules, and updates dependent fields.
- **Communication Layer:** The platform email engine formats and dispatches the acknowledgment message directly to the requester.
- **Governance & Security Layer:** ServiceNow Access Control Lists (ACLs), role definitions, and Update Sets safeguard schema integrity.

Architecture Components

Component	Role in the solution	Technology
User Interface	Students enter Caller and Short Description and access School IT Tickets.	ServiceNow Next Experience
Ticket Data	Stores Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group and Assigned To.	Custom table: u_school_it_ticket
Automation Logic	Checks the Short Description and updates Category and Subcategory for supported issue types.	ServiceNow Flow Designer
Notification	Sends an acknowledgment email to the Caller after ticket submission.	ServiceNow Email / SMTP
Security	Controls access to the custom table and its records.	ServiceNow ACLs and Roles
Change Tracking	Captures custom table definitions, choice fields, flow logic, and email configurations.	ServiceNow Update Sets

Security & Governance Architecture

The solution incorporates security and governance through platform-native controls: Access Control Lists (ACLs) prevent unauthorized record deletion or modification, role-based access restricts configuration capabilities to system administrators, and all changes are strictly encapsulated within an Update Set.

Solution Architecture - Implementation Evidence

The screenshot shows the 'School IT Ticket' form in ServiceNow. The form includes fields for Number (TKT0001012), Caller (System Administrator), Short description (Wi-Fi not working in library), and Description. There are dropdown menus for Category (Network), Subcategory (Wi-Fi), and State (None). There are also fields for Assigned Group and Assigned to. At the bottom, there are 'Update' and 'Delete' buttons.

Figure 2: School IT Ticket form showing user interface and entry fields.

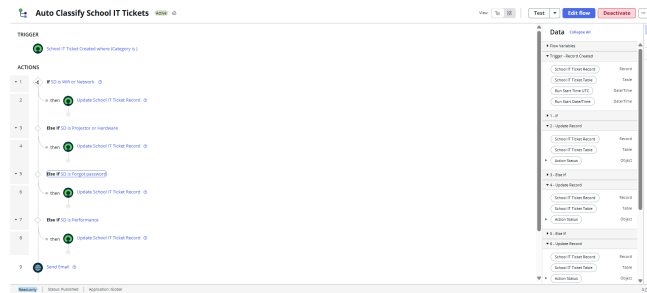


Figure 3: Published Auto Classify School IT Tickets Flow Designer workflow.

The screenshot shows the 'School IT Ticket' form in ServiceNow. The form includes fields for Number (TKT0001012), Caller (System Administrator), Short description (Wi-Fi not working in library), and Description. There are dropdown menus for Category (Network), Subcategory (Wi-Fi), and State (None). There are also fields for Assigned Group and Assigned to. At the bottom, there are 'Update' and 'Delete' buttons.

Figure 4: School IT Ticket showing Category = Network and Subcategory = Wi-Fi.

The screenshot shows the 'Completed Project Update Set' in ServiceNow. It includes a table with columns: Created, Type, User, Target name, Update type, Reason for update, and Action. The table contains several rows of data, including updates for 'Application Menu', 'Project', 'Checklist', 'Checklist', 'Checklist', 'Task', and 'Checklist'. Below the table, there is a section for 'Related Links' and a table for 'Update Set History'.

Created	Type	User	Target name	Update type	Reason for update	Action
2020-09-28 10:41:32	Application Menu	School IT Support	School IT Support	insert	INSERT_OR_REPLACE	
2020-09-28 10:41:32	Project	School IT Ticket	School IT Ticket	insert	INSERT_OR_REPLACE	
2020-09-28 10:41:32	Checklist	School IT Ticket Category	School IT Ticket Category	insert	INSERT_OR_REPLACE	
2020-09-28 10:41:32	Checklist	School IT Ticket Rules	School IT Ticket Rules	insert	INSERT_OR_REPLACE	
2020-09-28 10:41:32	Checklist	School IT Ticket Subcategory	School IT Ticket Subcategory	insert	INSERT_OR_REPLACE	
2020-09-28 10:41:32	Task	School IT Ticket	School IT Ticket	insert	INSERT_OR_REPLACE	
2020-09-28 10:41:32	Checklist	School IT Ticket	School IT Ticket	insert	INSERT_OR_REPLACE	
2020-09-28 10:41:32	Checklist	School IT Ticket Assigned Group	School IT Ticket Assigned Group	insert	INSERT_OR_REPLACE	

Figure 5: Completed Project Update Set with captured customer updates and XML export link.